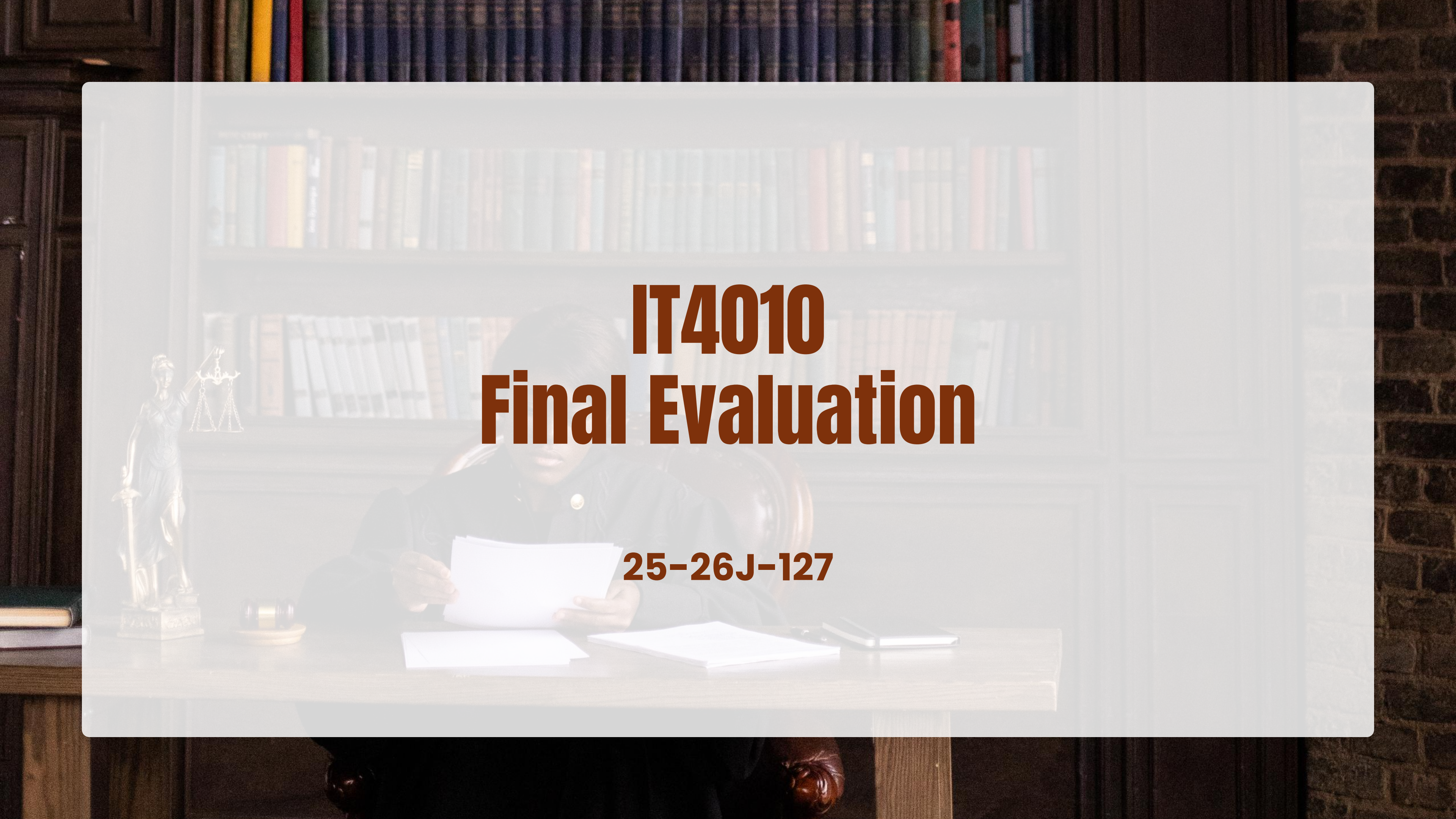




LEGALVISION

*A Knowledge-Driven AI Framework for Legal
Understanding and Trust Assessment*



IT4010 Final Evaluation

25-26J-127

PP2 COMMENTS & ACTIONS TAKEN

IT number	PP2 Comment	Actions Taken
IT22732736	Validate the model whether nodes are being formed correctly	Validated Neo4j node and relationship creation using sample deed records. Checked key nodes such as Instrument, Party, Property, Plan, Statute, Legal Principle, and Requirement. Verified relationship mapping and improved graph consistency.
IT22348784	Try to compare with legalbert along with RAG based system	Reviewed LegalBERT as an alternative model and compared it with the proposed LLaMA-based RAG approach. Since our module requires explainable answer generation rather than only text classification, the fine-tuned LLaMA model with Neo4j-based RAG was selected as the most suitable approach.
IT22113504	Try to compare your appraoch with a llm	Compared with Chatgpt and gemini • Our System (Correct), • ChatGPT (Incomplete + Hallucination), and • Gemini (Hallucination)

A dark brown leather suitcase with a brass handle and a book titled 'THE LAW' with a scale of justice emblem.

PROBLEM OVERVIEW

Problem: Legal documents are complex and time-consuming to analyze. Users need an automated system to understand legal content without consulting legal professionals for minor tasks, while ensuring traceability.

SOLUTION OVERVIEW

Solution: An AI-powered platform for analyzing Sri Lankan property law documents through summarization, explainable reasoning, risk evaluation, and knowledge graph support.



VISUAL INTEGRATED MULTI PERSPECTIVE LEGAL SUMMARIZER

IT22113504 | WICRAMASINGHE D A T N

This component develops a system that simplifies complex property law documents by generating multi-perspective summaries with visualizations and traceability, improving understanding for both experts and non-experts.



EXPLAINABLE LEGAL REASONING MODULE

IT22348784 | SIVANUJA S

Explainable Legal Reasoning Module generates structured IRAC-based, citation-backed legal explanations for Sri Lankan property law, enabling accurate, transparent, and trustworthy guidance, outperforming general and legal LLMs in evaluation.



BIAS & RISK EVALUATOR

IT22344892 | MAXWELL L Y

Bias & Risk Evaluator analyzes property agreements using NLP to detect biased clauses and risk severity, generating a confidence-based Trust Score and interactive report for clearer, user-friendly legal risk assessment.



LEGAL KNOWLEDGE GRAPH

IT22732736 | SHARAN K

Dynamic Legal Knowledge Graph converts Sri Lankan deed PDFs into structured graph data using NER and hybrid extraction, enabling ownership tracking, encumbrance analysis, and natural-language legal queries through GraphRAG.

RESULTS & TESTIMONIALS

The system was successfully **accepted by our external supervisor & get future enhancement suggestions**

Additionally, this research has been accepted for publication at the **CICN Conference**

Limitations of the System – Currently limited to the property law domain only

Future Improvements – Expand system to support multiple legal domains, & Integrate AI-based validation for identifying forged or suspicious documents



THANK YOU